

Retail Sales Analysis Dashboard

~Exploratory Data Analysis (EDA) Summary Report

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Objective

The objective of this project was to analyze retail sales data and generate meaningful business insights using Exploratory Data Analysis (EDA). The project focused on cleaning the dataset, performing filtering and sorting operations, calculating sales and profit statistics, analyzing region-wise and category-wise performance, and creating visualizations to understand sales trends and business performance.

Dataset Information

Dataset Name: Superstore Sales Dataset

Source: Kaggle

Dataset Link:

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Dataset Description

The Superstore Sales dataset contains information about customer orders, product categories, sales, profit, regions, and other business-related details. It is widely used for sales analysis and business intelligence to understand customer purchasing patterns and evaluate business performance across different regions and product categories.

Data Exploration

The dataset was explored using functions such as `head()`, `shape`, `info()`, and `describe()`. These functions helped understand the dataset structure, identify data types, examine the number of records, and generate descriptive statistics before performing detailed analysis.

Data Cleaning

The dataset was checked for missing values and duplicate records. Missing values were handled appropriately, duplicate records were removed, and the dataset was verified for consistency. These preprocessing steps ensured that the data was clean, reliable, and ready for analysis.

Statistical Analysis

Descriptive statistical measures such as mean, median, mode, variance, and standard deviation were calculated for numerical columns including Sales and Profit. These measures provided a better understanding of the overall business performance and the variation in sales and profit values.

Filtering and Sorting

Filtering and sorting operations were performed to analyze specific regions, product categories, and customer segments. These operations helped compare business performance across different groups and identify areas with higher sales and profitability.

Sales and Profit Analysis

Region-wise and category-wise analysis was performed to evaluate business performance. Total sales and profit were calculated for different regions and product categories. This analysis helped identify the best-performing regions, profitable categories, and areas that require improvement.

Data Visualization

Several visualizations were created to understand business trends and sales performance. These included Bar Chart, Line Chart, Histogram, Scatter Plot, Correlation Heatmap, and Box Plot. The visualizations helped identify sales patterns, profit distribution, and relationships between different business variables.

Key Insights

The dataset was successfully cleaned before analysis by handling missing values and duplicate records. Statistical analysis provided valuable insights into sales and profit distribution. Region-wise and category-wise analysis highlighted the highest-performing

business segments. Visualizations clearly illustrated sales trends, customer purchasing patterns, and relationships between sales and profit.

Conclusion

This project provided practical experience in performing Exploratory Data Analysis using retail sales data. It covered data cleaning, filtering, sorting, statistical analysis, sales and profit analysis, and visualization techniques. The project demonstrated how EDA can be used to generate meaningful business insights and support data-driven decision-making.

Tools Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Colab
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Project Outcome

Through this project, I gained hands-on experience in:

- Data Cleaning
 - Exploratory Data Analysis (EDA)
 - Data Filtering and Sorting
 - Statistical Analysis
 - Sales and Profit Analysis
 - Business Insight Generation
 - Data Visualization
 - Interpreting Analytical Results
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